

Francisco López Alonso

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Summary — SWE with 6 years of professional experience specialized in the development of distributed systems.

Skills

Cloud GCP

Programming Expertise Golang, TypeScript & Python Languages English, French & Spanish

Professional Experience

Schneider Electric

Sept 2023 – Present
Nice, France

Software Engineer - Full Stack

Project Context: Automation Expert is an industrial platform that allows engineers to model device topologies (PLCs, HMIs) and build executable applications that run directly on physical devices. Applications are composed within a defined topology, consume input signals, produce outputs, and can be remotely operated through controlled lifecycle actions such as authentication, deployment, execution, and result inspection on real hardware.

Stack: C#, Rust, React and Python

- Designed a **state machine** to manage and execute operations on physical devices within the Automation Expert platform. For example, deploying an application to a device involves multiple steps, including authentication, with the deployment itself modeled as an action within the state machine to ensure correct sequencing and execution.
- Implemented **automation scripts** to ease the integration and deployment of microapps within the Automation Expert platform, significantly reducing manual developer overhead.
- Improved internal development velocity by standardizing microapp packaging and deployment flows.

Project Context: Edge platform for IoT devices providing an end-to-end system to collect data from PLC-connected devices, process it in near real time, and enable operational actions based on live device signals. The platform supports telemetry ingestion, event-driven alerting, and data lifecycle management across distributed industrial environments.

Stack: Golang, MQTT, EdgeX and eKuiper

- Designed and implemented an **event-driven alert system with MQTT** for real time monitoring of device variables, enhancing client visibility and control over device variables values.
- Designed and implemented 5 of 30 **React** microfrontends using **Single-SPA** and **TypeScript**, each responsible for distinct business domains (e.g computation and alerting dahsboards) enabling independent development without disrupting other modules.
- Collaborated with backend teams to integrate microservices via REST and GraphQL APIs, providing consistent data flows across **distributed frontends** within a unified UX.
- Built a **database retention and archiving mechanism** that automatically compresses and transfers aged IoT records to cold storage when the active dataset exceeds 10GB, reducing live DB footprint by over **30%**.
- Technical support to three developers.

Symphony Communications

Apr 2022 – Sep 2023
Sophia Antipolis, France

Software Engineer II - Frontend

Project Context: Secure financial messaging platform used by global banks to exchange regulated communications. The system operates as a modular monolith, handling message flows

Stack: TypeScript, React, Redux, Docker, BackboneJS and Single SPA

- Led the **migration from Backbone.js to React**, refactoring legacy components (like admin and audit trail dashboards) into reusable modules and enabling the frontend transition from a monolith to a modular architecture.
- Optimized build time performance from **Webpack** to **Vite** reducing initial load times and improving developer productivity across multiple frontends.
- Collaborated with backend and DevOps teams to **modularize build pipelines**, introducing GitHub Actions and targeted Jenkins jobs that significantly reduced deployment time.
- Integration of **Snyk vulnerability scanning** into CI/CD workflows.
- Enhanced the **shared UI component library** to support dynamic runtime loading, accelerating delivery of new features across teams.
- Worked within a **7-person cross-functional team**, contributing to product design decisions and frontend scalability improvements.

Cubica Comunicación

Jul 2021 – Mar 2022
Vigo, Spain

Software Developer - Full Stack
Project Context: Internal CRM for Xplora Riviera, a Mexican travel company, to manage trips, bookings, and customer data
Stack: TypeScript, React, Redux, MySQL and Grafana

- Delivered a **CRM application** in collaboration with the CTO and design team, improving booking efficiency and customer tracking through real-time dashboards.
- Integrated **automated invoicing and notification systems**, eliminating manual steps in the reservation workflow.
- Built **Grafana-based analytics dashboards** to visualize purchasing patterns, enabling data-driven business decisions.

Certifications

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| – Google Cloud Engineer Associate | Issued: 2025 |
| – Google Cloud Digital Leader | Issued: 2024 |
| – Symphony Certified Developer | Issued: 2022 |

Education

Double Major
- Bachelor of Computer Engineering at University of Vigo
Minors: Cloud Computing;
- Bachelor of Business Management at University of Vigo
Minors: Banking and Fractional Reserve;